

# Final Project: AI328

## Fraud Detection Using Financial Transactions

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### Problem Definition and Objectives:

Fraud detection is critical for financial institutions to ensure the security of transactions and protect users' accounts. Fraudulent activities can result in substantial financial and reputational losses. This project aims to develop a machine learning-based solution to identify fraudulent transactions effectively.

#### Objectives:

- **Primary Goal:** To detect fraudulent transactions from a dataset of financial transactions.
  - **Sub-Goals:**
    - Build a robust dataset through preprocessing.
    - Compare the performance of multiple models for detecting fraudulent transactions.
    - Derive actionable insights from the data through visualizations.
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## Data Description and Preprocessing Steps:

### Dataset Overview:

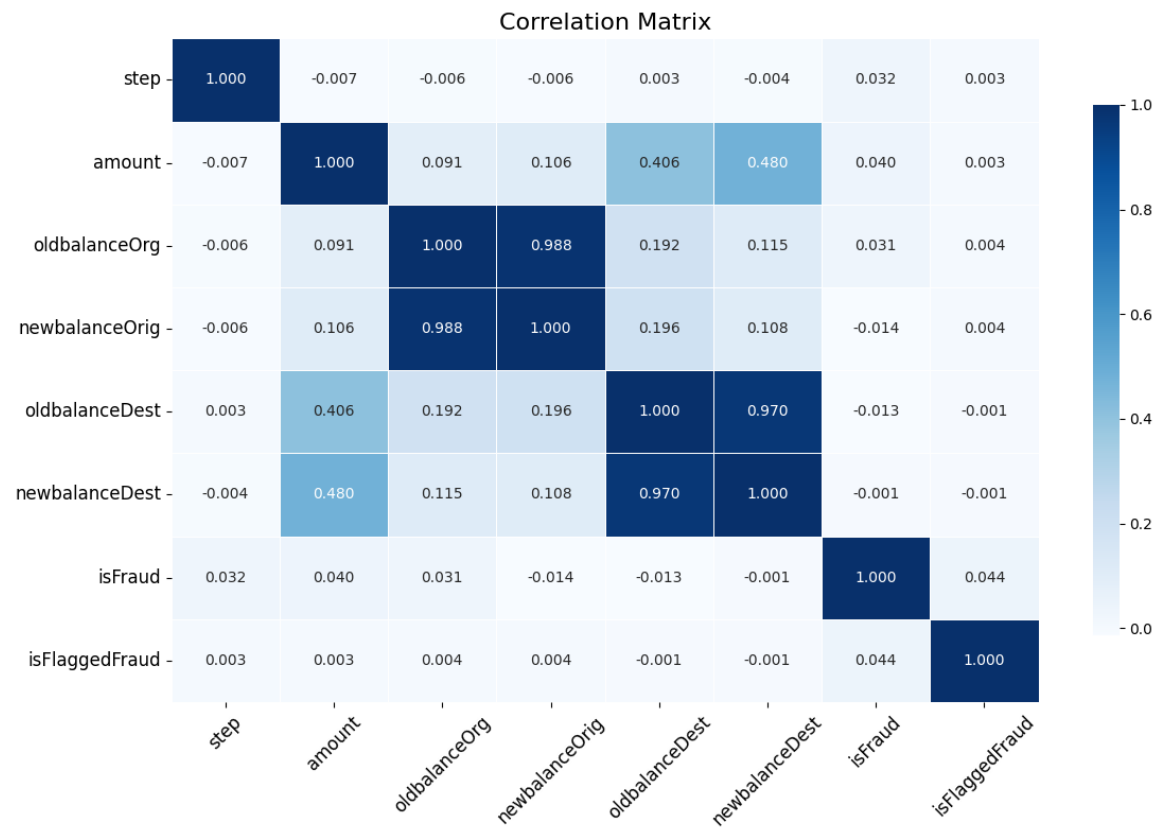
The dataset contains millions of financial transaction records. Key features include:

- **Step:** Time step of the transaction.
- **Type:** Type of transaction (e.g., CASH\_IN, CASH\_OUT, PAYMENT, TRANSFER, etc.).
- **Amount:** The amount of the transaction.
- **OldbalanceOrig:** Sender's balance before the transaction.
- **NewbalanceOrig:** Sender's balance after the transaction.
- **OldbalanceDest:** Receiver's balance before the transaction.
- **NewbalanceDest:** Receiver's balance after the transaction.
- **isFraud:** Label indicating whether the transaction was fraudulent.

### Preprocessing Steps:

- **Handling Missing Values:** No missing values were present in the dataset. However, irrelevant columns were checked and excluded if necessary.
- **Feature Engineering:**
  - **One-Hot Encoding:** Applied to the categorical `Type` column, converting transaction types into numerical format.
  - **Scaling:** The numeric columns (e.g., `Amount`, `OldbalanceOrig`) were scaled using standardization to enhance model convergence.
- **Data Balancing:**
  - The dataset was highly imbalanced (only 0.1% fraudulent transactions). To address this:
    - **Undersampling:** Non-fraudulent transactions were reduced to balance the classes.

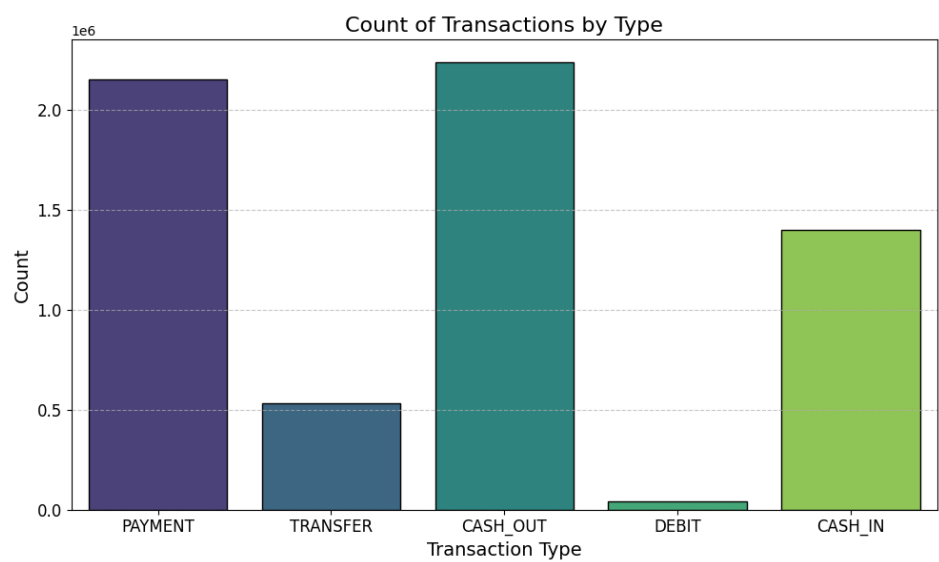
- The resulting dataset had equal samples for fraudulent and non-fraudulent transactions.
- **Correlation Analysis:**
  - A heatmap of feature correlations was generated to analyze relationships. The highest correlations were observed between balances and amounts, indicating their predictive significance.
  - **Figure 1: Correlation Matrix** was included to support this analysis.



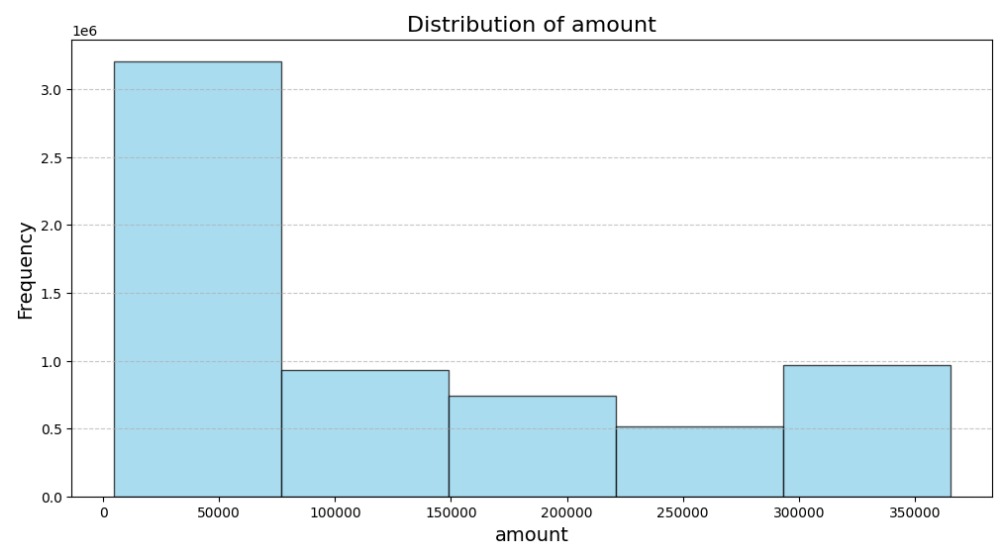
- **Feature Selection:**
  - Removed redundant or irrelevant features based on low correlation or business logic (e.g., derived variables with minimal variance).

**Key Preprocessing Visualizations:**

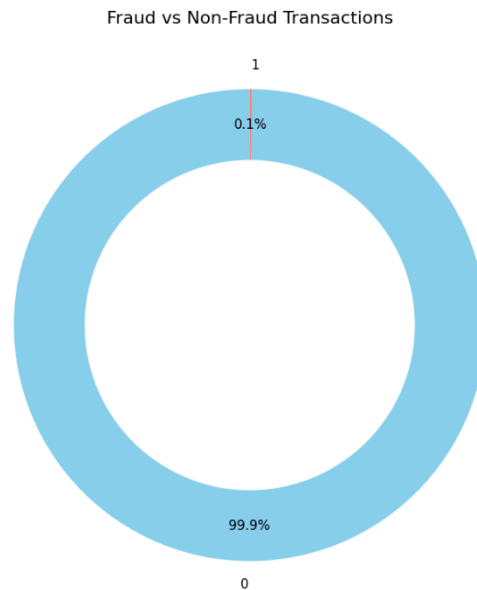
**Distribution of Transaction Types:**



**Distribution of Amounts:**



## Fraud vs Non-Fraud Distribution:



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## Model Selection, Training, and Evaluation

### Model Selection

Four models were selected for comparison:

- **Logistic Regression (LR):** A baseline for binary classification.
- **XGBoost (XGB):** A tree-based boosting model, well-suited for imbalanced data.
- **Naïve Bayes:** A probabilistic model.
- **Support Vector Classifier (SVC):** Effective for linear and non-linear separable data.

## Training Process

- **Dataset Splitting:** The data was split into 70% training and 30% testing datasets.
- **Hyperparameter Tuning:** Grid search and random search were used to optimize model parameters.

## Evaluation Metrics

Models were evaluated using:

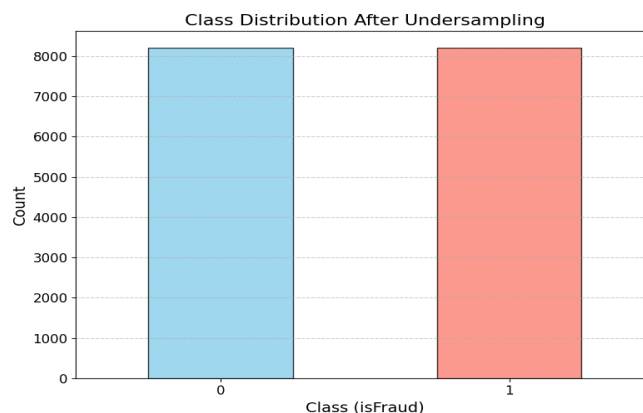
- **Accuracy:** Overall correct predictions.
- **Precision:** Focus on true positives.
- **Recall:** Coverage of fraudulent transactions.
- **F1 Score:** A harmonic mean of precision and recall.
- **F-Beta ( $\beta=0.5$ ):** Adjusted precision-recall balance.

## Results:

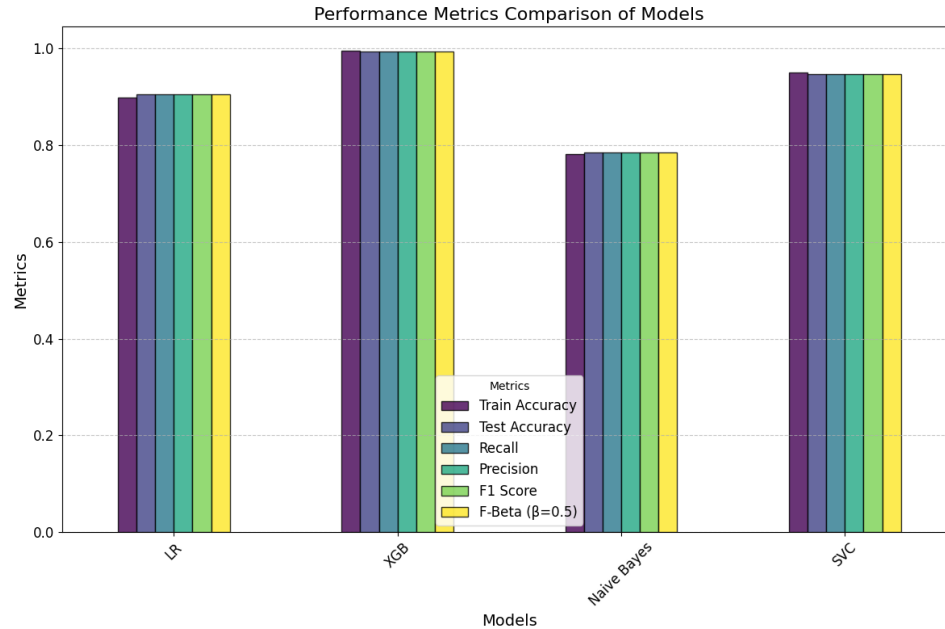
- **Logistic Regression (LR):** Test Accuracy: 90.4%, Precision: 90.4%.
- **XGBoost (XGB):** Test Accuracy: 99.4%, Precision: 99.4%.
- **Naïve Bayes:** Test Accuracy: 78.4%, Precision: 78.4%.
- **SVC:** Test Accuracy: 94.7%, Precision: 94.7%.

## Key Model Visualizations:

**Class Distribution After Balancing:**



## Performance Metrics Comparison of Models:

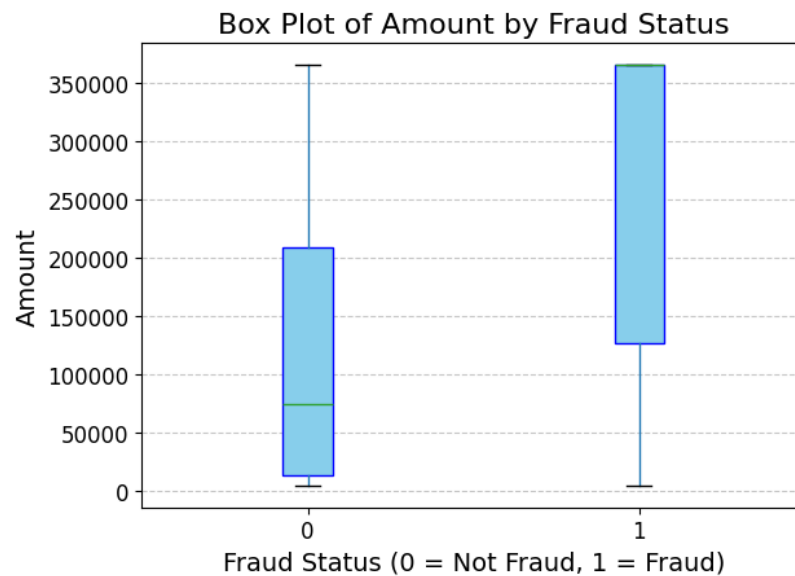


## Results and Insights

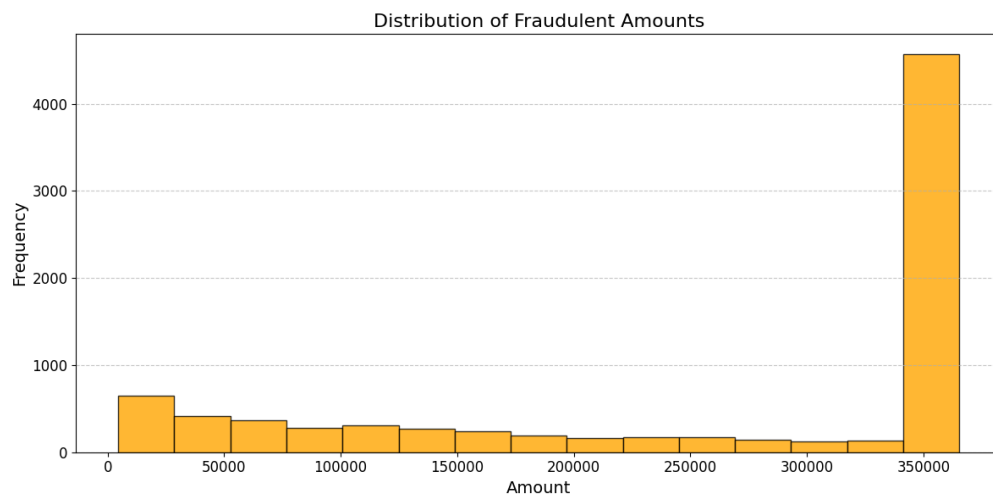
- **Fraudulent Transactions:**
  - Fraudulent transactions are predominantly observed in TRANSFER and CASH\_OUT transaction types.
  - The highest amounts are generally associated with fraudulent transactions.
- **Feature Importance:**
  - The most predictive features are amount, oldbalanceOrg, and newbalanceOrig.
- **Challenges in Detection:**
  - Imbalanced datasets increase false-negative rates.
  - Certain features show minimal variance or misleading trends.
- **Top Model Performance:**
  - **XGBoost** outperformed all other models with a near-perfect accuracy of 99.4%.

## Key Visualizations:

### Box Plot of Amount by Fraud Status:



### Fraudulent Transaction Distribution:



## Challenges Faced

1. **Data Imbalance:** The extreme class imbalance required innovative balancing techniques, which were computationally intensive.
2. **Scalability:** Processing millions of records demanded optimization of code and computational resources.



- 3. Model Overfitting:** Certain models, like SVC, exhibited overfitting to the training data, necessitating regularization.

[This code source](#)